

7.4.1.13 Service channel error messages

Should an error occur in the transport mechanism of the service channel (e.g., if the lengths of the operation data differ between the master and the slave, or vice versa, or if the IDN is undefined), the slave shall announce it by setting the error bit (bit 2) in the status word and by writing an error code into the service INFO field of its AT.

The slave shall be allowed to report an error message only if a new processing step is issued by the master, which is in any of following cases:

- MHS-bit $\neq$ AHS-bit (step not yet secured);
- busy bit = 1 (step still in process).

If the slave recognizes an error, it shall ignore the actual step, interrupt and acknowledge by:

- setting the AHS-bit equal to the MHS-bit (if not already acknowledged in a previous cycle);
- setting the error bit to 1 (SVC status, bit 2);
- setting the busy bit to 0;
- setting the SVC valid to 1;
- sending the error codes in the AT SVC INFO field (see AT SVC INFO).

If the master intends to start a transmission of an element after an error message, the slave shall update the SVC status and AT SVC INFO.

All possible SVC error messages are shown in Table 96.

Table 96: Error messages

Error code	Description	Comment
0x0nnn	General error	
0x0000	No error in the service channel	
0x0001	Service channel not open	
0x0009	Invalid access to closing the service channel	
0x1nnn	Element 1	Identification number
0x1001	IDN not supported	
0x1009	Invalid access to element 1	
0x2nnn	Element 2	Name
0x2001	Name not supported	
0x2002	Name transmission too short	Master set "last transmission" too early
0x2003	Name transmission too long	Master does not set "last transmission"
0x2004	Name cannot be changed	Name is read only
0x2005	Name is write-protected at this time	

0x3nnn	Element 3	Attribute
0x3002	Attribute transmission too short	Master set "last transmission" too early
0x3003	Attribute transmission too long	Master does not set "last transmission"
0x3004	Attribute cannot be changed	Attribute is read only
0x3005	Attribute is write-protected at this time	
0x4nnn	Element 4	Unit
0x4001	Unit not supported	
0x4002	Unit transmission too short	Master set "last transmission" too early
0x4003	Unit transmission too long	Master does not set "last transmission"
0x4004	Unit cannot be changed	Unit is read only
0x4005	Unit is write-protected at this time	
0x5nnn	Element 5	Minimum input value
0x5001	Minimum input value not supported	
0x5002	Minimum input value transmission too short	Master set "last transmission" too early
0x5003	Minimum input value transmission too long	Master does not set "last transmission"
0x5004	Minimum input value cannot be changed	Minimum input value is read only
0x5005	Minimum input value is write-protected at this time	
0x6nnn	Element 6	Maximum input value
0x6001	Maximum input value not supported	
0x6002	Maximum input value transmission too short	Master set "last transmission" too early
0x6003	Maximum input value transmission too long	Master does not set "last transmission"
0x6004	Maximum input value cannot be changed	Maximum input value is read only
0x6005	Maximum input value is write-protected at this time	
0x7nnn	Element 7	Operation data
0x7002	Operation data transmission too short	Master set "last transmission" too early
0x7003	Operation data transmission too long	Master does not set "last transmission"
0x7004	Operation data cannot be changed	Operation data is read only
0x7005	Operation data is write-protected at this communication phase	
0x7006	Operation data is smaller than the minimum input value	
0x7007	Operation data is greater than the maximum input value	

0x7008	Invalid operation data	which may be an unsupported • bit number or bit combination, • value, code or • configured IDN
0x7009	Operation data write protected by a password	
0x700A	Operation data is write protected, it is configured cyclically	IDN is configured in the MDT or AT. Therefore writing via the service channel is not allowed.
0x700B	Invalid indirect addressing	e.g., data container, list handling etc.
0x700C	Operation data is write protected, due to other settings	e.g., operation mode, sub-device is enabled, setting of communication version etc.
0x700D	Invalid floating point number	
0x700E	Operation data is write protected at parameterization level	
0x700F	Operation data is write protected at operating level	
0x7010	Procedure command already active	
0x7011	Procedure command not interruptible	
0x7012	Procedure command at this time not executable	e.g., in this phase the procedure command can not be activated.
0x7013	Procedure command not executable	the corresponding parameters are invalid or false
0x7014	The received current length of list parameter does not match to expectation	
0x7015	Operation data is not yet created completely	it takes more time to create the operation data, try it again later
0x71nn	Segment wise transmission of list parameters via SVC	
0x7101	IDN in S-0-0394 not valid	
0x7102	Empty list in S-0-0397 not allowed for write access	
0x7103	Maximum length of the list in S -0-0394 is exceeded by take-over of the list segment.	
0x7104	Read access only: The length of the list segment as of the list index exceeds the current length of the list in S-0-0394.	
0x7105	IDN in S-0-0394 is write protected	

0x7106	Operation data in list segment is smaller than the minimum input value	
0x7107	Operation data in list segment is greater than the maximum input value	
0x7108	Invalid list index in S-0-0395	
0x7109	Parameter in IDN S-0-0394 does not have variable length	
0x710A	IDN S-0-0397 not permitted as operation data in S-0-0394	
0x8nnn	Reserved for master internal error codes	Error codes may be defined by the manufacturer of control units (e.g. NC, PLC etc.)
0xAxxx	Reserved	
0xBxxx	Reserved	
0xCxxx	Reserved for slave specific error codes	used for fault analysis and trace functionality (troubleshooting)
0xDxxx	Error codes are not generated and transmitted via SVC	Error codes shall be defined by a TWG of sercos
0xD000	no error	
0xD001	service channel (temporarily) not available	
0xD002	service channel engaged by an application	
0xD003	service channel busy, slave is processing previous request	
0xD004	sercos Slave not reachable	
0xD005	service channel transaction aborted	
0xD006	writing this element is not supported by the service channel	
0xExxx	Reserved for master internal error codes	Error codes may be defined by the manufacturer of control units (e.g. NC, PLC etc.)
0xFxxx	Reserved for master internal error codes	Error codes may be defined by the manufacturer of control units (e.g. NC, PLC etc.)
All other error codes shall be reserved.		